Pushing POLYGONS

The Parhelia and the P10 will redefine the technologies used for 3D graphics. Enter the age of super realism where the differences between the surreal and the real world will blur away

The past few years have seen some dramatic changes in the 3D graphics industry: the fall of a monopoly (3dfx), the rise of another (nVidia), the evolution of the graphics processor, and the rise of a contender for the top spot in the form of ATi. It has been an interesting ride for sure. But all along, the consumer has had little choice in what he could buy, and how much he had to pay for it. It was the economics of competition, or in this case the lack of competition, that had forced such a situation upon us.

But things seem to be changing. Competition as they say, is coming out of the woodwork. Companies long thought dead and forgotten have risen to lay claim to a slice of the market pie, and things will only get better. What is it about the current scenario that has companies feeling confident enough to take on the likes of nVidia and ATi?

Standardisation is the magic word. Thanks to the introduction of Application Programming Interfaces (APIs) such as DirectX 8 and the inevitable arrival of DirectX 9 and OpenGL 2.0, the playing field has been greatly levelled. By standardising what should go inside a 3D chip, smaller companies no longer have to worry whether the features they've introduced will be used by third-party developers. They can now concentrate their efforts on API compatibility, even as they innovate in technologies that might give them an edge such as in terms of the product cost (a cheap yet powerful graphics chip like the Kyro II), or the market positioning of the product (a part designed for the notebook segment, for example).

We thus have solutions being offered by old favourites like Matrox and 3Dlabs, and a few surprise entrants like SiS and VIA. The more interesting of these are the upcoming chips from Matrox and 3Dlabs, if only because they are aimed at the high-end market, where the current giants (read nVidia and ATi) have a seemingly unbreakable hold.

Powerfully Matrox

Think 2D quality, and a Matrox card comes to mind. For years Matrox has been the benchmark against which the 2D quality of any card has been measured. Add to that their DualHead technology, which allows a user to use two monitors simultaneously, and it's easy to see why Matrox has been a perennial favourite. For those who are sceptical of Matrox's ability to deliver a competitive 3D card, they have one word: Parhelia. Built using cutting-edge 0.15-micron manufacturing process and powered by a whopping 80 million transistors and a 512-bit architecture, the Parhelia-512 promises a new level of visual quality and blazing next-generation performance. And it is aimed straight at 3D gamers.

The right stuff

The Parhelia has its memory running at a staggering 325 MHz DDR (effectively 650 MHz). Add to this the 256-bit-wide memory interface, and you get an effective bandwidth of 20.8 GBps. Not impressed? Compare this with the 8.8 GBps bandwidth of the ATi Radeon 8500 and the 10.4 GBps boasted by the nVidia GeForce4 Ti 4600, and you can easily see that the Parhelia has the power to make light work of just about any 3D scene. However, the chip is not DirectX 9 compliant. While it does support DirectX 9 vertex shaders, it is limited to v1.3 pixel shaders, and is thus not a true DirectX 9 part.

Of pixel and vertex shaders

The Parhelia integrates four Microsoft DirectX 9 compatible vertex shader units into a single vertex processing array and is able to sustain very high performance lev-